

City University of Hong Kong 2025-2026
Semester A CS3343 Software Engineering
Practice

Project Plan
Group 19
Classic UNO Card Game

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1. Summary	3
1.1 Project Description	3
1.2 Stakeholders	3
1.3 Objectives of the Project	3
1.4 Project Requirements	4
2. Software Development Methodology	5
2.1 Project Team Organization	5
2.2 Development Tools	5
2.2.1 Development IDE Software	5
2.2.2 Development Platform	5
2.2.3 Test Cases	5
2.2.4 Coverage Checking	6
2.2.5 Testing Platform	6
2.2.6 Documentation	6
2.2.7 Version Control	6

1. Summary

1.1 Project Description

UNO has been a well known card game around the globe for years. The simple game rules and fun, interactive gameplay are the main reasons why it has been so popular since its release. However, some players may have difficulty finding friends and suitable locations to play UNO.

Therefore, our group has developed a virtual version of UNO, which allows players to play UNO with CPU Players anywhere they want. This way, they can enjoy the game they love whenever they want, regardless of the settings.

1.2 Stakeholders

Stakeholder	Description
User	The users are the ones using our application to play UNO. They can also provide feedback for the developers to improve the game
Developers	The developers are the team members responsible for maintaining the operation of the game system and launching future updates

1.3 Objectives of the Project

The main objectives of the project are:

- **Virtual Version of UNO**

The main objective of this project is to develop a virtual version of UNO. We believe there is a demand for a UNO that players can play anywhere regardless of their location.

- **Convenient and User Friendly**

We also put emphasis on optimising user experience. We achieve this by providing a clean and easy-to-navigate user interface. Users can start a new game, play cards, and use the functions in our game without facing any confusion or difficulty.

- **Single Player Viability**

We understand that some people may have trouble finding other peers to play UNO with, therefore, our game is for single player, and players can play UNO with CPU players. Not only do they get to enjoy UNO, they can also try to analyse the playing patterns of CPU players, hence adding another layer of challenge into the game.

1.4 Project Requirements

The functional requirement(s) in this project include:

- 1. Start new game**

Once players open our game, they will be directed to the home page, where they can start a new game, playing with CPU players.

- 2. All available actions in UNO**

During the game, users are able to perform actions in a UNO game such as:

Playing a Card

Drawing Card(s)

Calling “UNO”

Catching Other Players Who Forgot to Call UNO, etc.

- 3. CPU Players**

Players will be playing with CPU players with fixed playing logic.

The non-functional requirement(s) in this project include:

- 1. User Friendly Interface**

To optimize playing experience, the game provides an organised user interface for users to play the game of UNO without any inconvenience or difficulties regarding navigating the app.

2. Software Development Methodology

When developing this UNO project, a divide-and-conquer approach was used. In the early stages of the project, we defined the requirements and corresponding functional components of the game, such as playing cards, calling "UNO", etc. Our group then divided up the components for each team member to develop and test independently. After that, we integrated the code and improved code quality over time via refactoring and future updates.

2.1 Project Team Organization

Name	Title	Description
Woo King Hei	Assistant Project Manager	Testing

2.2 Development Tools

2.2.1 Development IDE Software

- Eclipse IDE for Java Developers - 2024-09
- Visual Code Studio

2.2.2 Development Platform

- Windows 10 with Java Development Kit (JDK) v16.0.2

- macOS Monterey 12.0.1 (x86_64)

2.2.3 Test Cases

- JUnit v5

2.2.4 Coverage Checking

- Eclipse IDE built-in coverage runner

2.2.5 Testing Platform

- Windows 10
- macOS Monterey 12.0.1 (x86_64)

2.2.6 Documentation

- Visual Paradigm 16.2 - Use Case Diagram, Class Diagram and Sequence Diagram
- Bugzilla - Bug Report
- Google Doc - Report

2.2.7 Version Control

Our group used Github for collaboration and version control. Github enables our team members to develop the code independently on their respective branches without affecting the main branch. Using Github also helped our team to track progress and changes due to updates. Overall, github greatly boosted our collaboration and efficiency when developing the code.

